Heart Disease Prediction

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Abstract

Heart disease is one of the leading causes of death worldwide. Early prediction can significantly improve treatment outcomes. This project develops a robust machine learning model using ensemble techniques to predict heart disease risk. The model achieves high performance with an ROC-AUC of approximately 0.956 and demonstrates strong generalization across cross-validation folds.

1 Introduction

Heart disease continues to be a major global health concern. Accurate early prediction systems can assist clinicians in diagnosis and treatment planning. This project focuses on building a high-performance machine learning model using clinical data to predict heart disease.

2 Methodology

2.1 Data Preprocessing

The dataset was cleaned and preprocessed by handling missing values, encoding categorical variables, and applying feature scaling where necessary.

2.2 Model Development

The following models were trained:

- Scikit-learn based ensemble model
- XGBoost model
- Multi-model ensemble (3 models per fold)

A 5-fold cross-validation strategy was used to ensure robustness.

3 Evaluation Metrics

The models were evaluated using:

- ROC-AUC (Receiver Operating Characteristic - Area Under Curve)
- PR-AUC (Precision-Recall Area Under Curve)
- Out-of-Fold (OOF) predictions

4 Results

4.1 Ensemble Model Performance

- Overall OOF AUC: 0.955745
- Mean Fold AUC: 0.955746
- Standard Deviation: 0.000463

Fold	AUC
1	0.956139
2	0.955002
3	0.955827
4	0.955480
5	0.956281

Table 1: Fold-wise AUC Scores

4.2 Model Comparison

4.2.1 Scikit-learn Model

- OOF AUC: 0.952519
- PR-AUC: 0.945471
- Std Dev: 0.000345

4.2.2 XGBoost Model

- OOF AUC: 0.955341
- PR-AUC: 0.948660
- Std Dev: 0.000432

4.2.3 Final Ensemble Model

- OOF AUC: 0.955745 (Best Performance)

5 ROC Curve Analysis

The ROC curve illustrates the trade-off between true positive rate and false positive rate. The model achieves an AUC of approximately 0.956, indicating excellent class separability.

6 Precision-Recall Curve Analysis

The Precision-Recall curve shows strong performance with PR-AUC close to 0.95, indicating high precision and recall. This is particularly important for medical datasets where class imbalance is common.

7 Out-of-Fold Predictions

OOF predictions provide an unbiased estimate of model performance. The results indicate:

- Good calibration
- Minimal overfitting
- Consistent performance across folds

8 Discussion

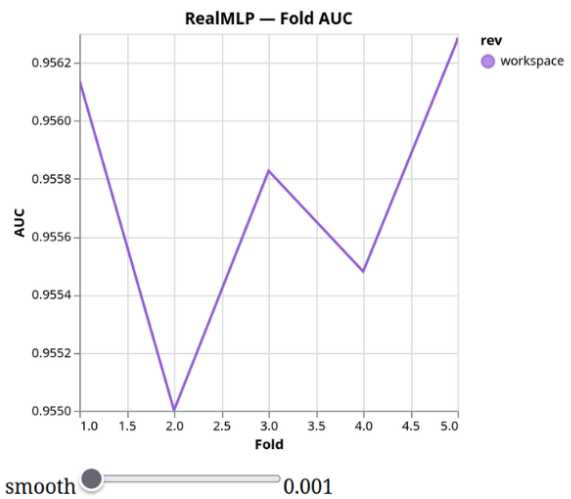
The ensemble model outperforms individual models due to improved generalization. XGBoost also performs strongly, confirming its effectiveness on structured data. The low standard deviation across folds indicates stable model performance.

9 Conclusion

This project successfully developed a high-performing heart disease prediction model with:

- ROC-AUC ≈ 0.956
- PR-AUC ≈ 0.95
- High stability across folds

The model is accurate, reliable, and suitable for real-world applications.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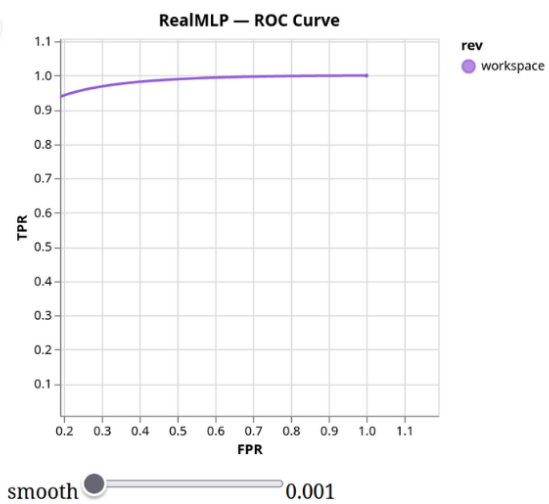
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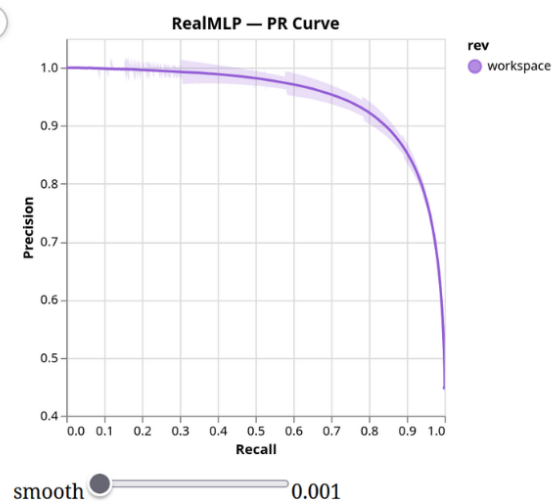
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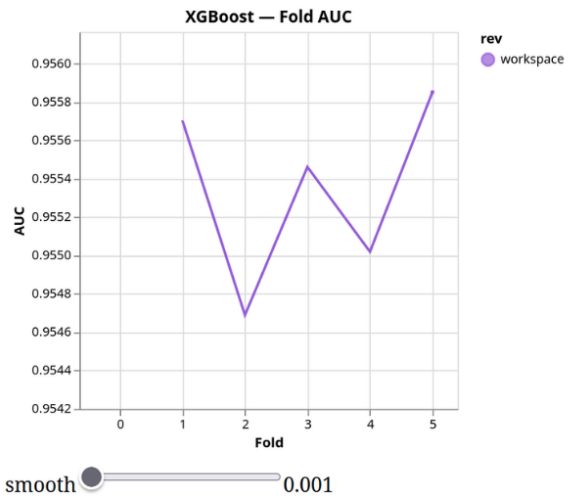
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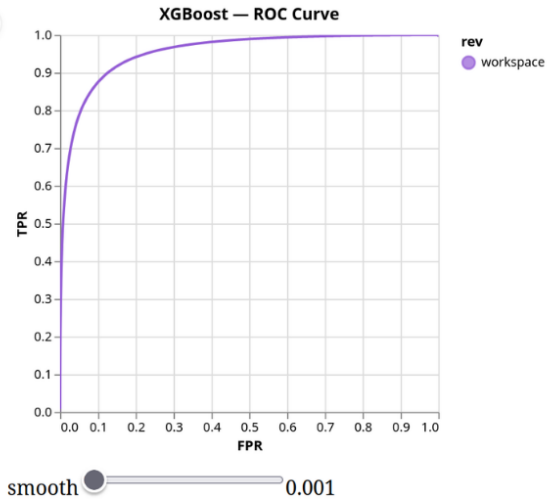
XGBoost — ROC Curve
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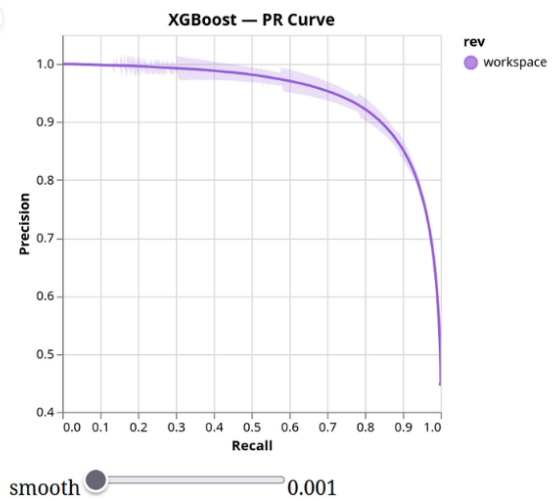
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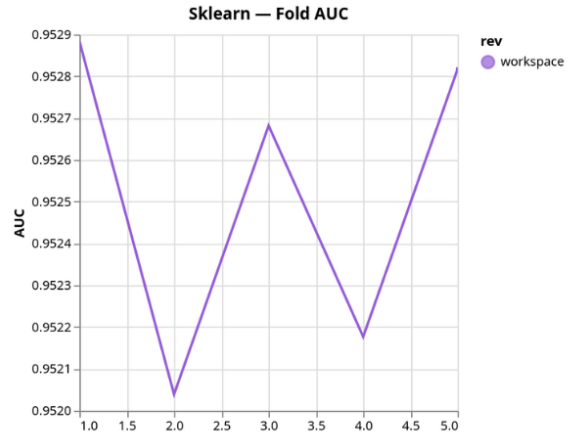
Sklearn — Fold AUC
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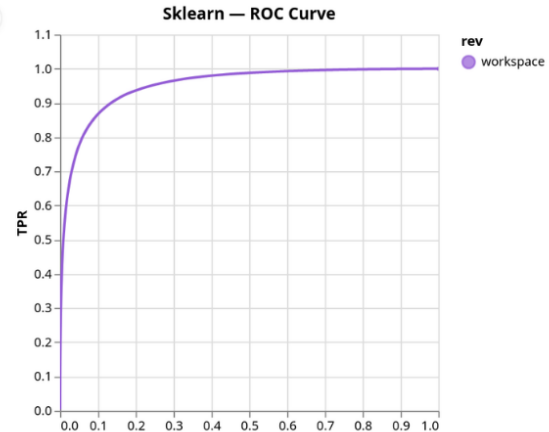
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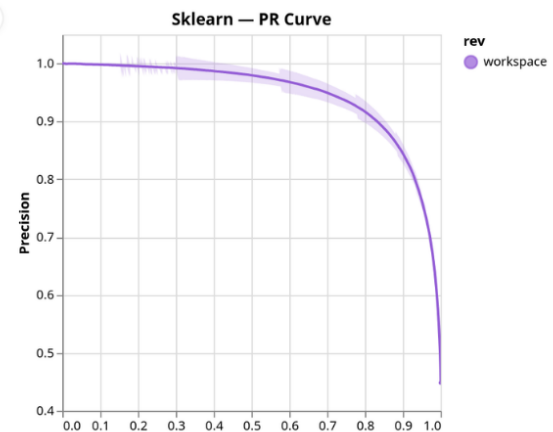
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